

Phase 6

Project Conclusion, Critical Appraisal, Forward Research Paths, and Frozen Record

Fixed structural reference: GRCh37/hg19 chr17:1,411,408-1,841,103 | heterozygous deletion | nominal span 429,696 bp | 17 affected annotated loci | INPP5K and RTN4RL1 boundary-intersected

1. Purpose and Phase 6 Status

Phase 6 concludes the first-pass in silico research program. It does not reopen or silently revise the frozen findings of Phases 1-5. Its role is to consolidate what the project established, identify what remains unsupported or unresolved, critically assess the methodology and inferential limits, define two legitimate continuation paths, and establish a version-controlled stopping point from which future work can proceed.

The project began with a broad visual-neurological question, including possible photosensitivity and visually triggered seizure mechanisms. The accumulated evidence narrowed that question. The strongest defensible interpretation is now a distributed visual-neural functional-reserve model involving retinal/RPE support, membrane and phosphoinositide biology, endosomal/lysosomal trafficking, genome-stress response, neural development, and network organization. The exact heterozygous CNV has not been shown to cause abnormal visual-cortical flicker processing, a photoparoxysmal response, or visually triggered seizures.

2. Project Facts Preserved at Freeze

Structural reference remains GRCh37/hg19 chr17:1,411,408-1,841,103, nominally 429,696 bp, heterozygous.

The interval affects at least 17 annotated loci: 13 protein-coding loci and 4 noncoding loci.

Eleven protein-coding genes are fully encompassed: PITPNA, SLC43A2, SCARF1, RILP, PRPF8, TLCD2, WDR81, SERPINF2, SERPINF1, SMYD4, and RPA1.

INPP5K and RTN4RL1 are boundary-intersected. Under the reported-coordinate model, approximately 49.6% of INPP5K MANE amino-acid-coding sequence and approximately 99% of RTN4RL1 CCDS sequence on the affected allele lie within the deletion.

The noncoding loci PITPNA-AS1, MIR22HG, MIR22, and RN7SL105P are included in the frozen interval.

The reported array coordinates are operational CNV boundaries, not demonstrated nucleotide-exact chromosome junctions.

3. Integrated Findings of the First-Pass Research

3.1 Structural genomics

Phase 1 converted a generic 17p13.3 label into a reproducible locus-specific substrate. The main structural result was that apparent genomic-span overlap can substantially understate coding disruption at the two boundary genes. The project therefore correctly anchored all later interpretation to the exact reported interval rather than importing phenotypes from broader 17p13.3 deletion syndromes.

3.2 Molecular and biochemical architecture

Phase 2 showed that the interval is not biologically featureless. Multiple affected genes occupy mechanistically relevant systems: phosphoinositide metabolism, phospholipid transfer and membrane composition, late-endosome and lysosome biology, intracellular transport, retinal trophic support, RNA splicing, DNA replication and repair, neuronal structural signaling, amino-acid transport, immune clearance, and fibrinolysis. The strongest mechanistic convergence involved INPP5K, PITPNA, TLCD2, WDR81, RILP, and SERPINF1, with RTN4RL1 adding

neuronal surface and structural signaling. This was evidence of functional adjacency and convergence, not proof of multigene synergy or epistasis.

3.3 Spatial and developmental opportunity

Phase 3 established where affected mechanisms could plausibly operate. The clearest direct cell-localized convergence was RPE-centered, particularly INPP5K, PITPNA, and SERPINF1. PRPF8 showed broad retinal relevance, and other loci provided distributed retinal, neuronal, cortical, and developmental opportunity. The negative results were equally informative: the affected interval did not emerge as a preferential magnocellular program in human dLGN data, and adult/developmental cortical analyses did not support a simple V1-specific expression model. Expression was explicitly treated as opportunity rather than dysfunction.

3.4 Dosage, perturbation, and phenotype-level evidence

Phase 4 moved beyond expression to perturbational evidence and genotype-state discipline. RPA1 provided the clearest direct human haploinsufficiency precedent; SERPINF1 showed direct heterozygous molecular dosage reduction of PEDF; INPP5K provided one-copy physiological evidence outside the retina together with strong phosphoinositide/lysosomal mechanism. Other loci supplied strong but less genotype-equivalent evidence from biallelic loss, knockout, hypomorphs, or variant-specific disease mechanisms. The project repeatedly preserved the distinction between severe loss and heterozygous deletion.

At the systems level, the retinal evidence favored impaired maintenance, reduced resilience, delayed recovery, or stress vulnerability more strongly than increased sensory gain. The best partial-dose retinal physiology identified in the project was reduced, not amplified, electrophysiological response. Likewise, RTN4RL1 complete-loss neural models altered network

construction but did not support a generic hyperexcitability narrative. MIR22 remained relevant to epileptogenesis, but its strongest evidence came from complete loss after an epileptogenic insult, not from heterozygous loss under photic stimulation.

3.5 Phase 5 deterministic integration and simulation

Phase 5 formalized the earlier record into a layered evidence model and treated missing downstream physiology as missing rather than as zero or as implied support. Its modeling and simulation work did not create biological evidence; it examined methodological behavior. The standardized simulation suite used 20,000 replicates per scenario/cell where applicable, approximately 1.88 million primary stochastic draws/resamples. It showed that modest heterozygous effects can be difficult to detect with too few independent biological replicates, that pseudoreplication cannot substitute for independent clones or donors, that broad frequency scans can generate false-positive signals without correction, and that limited data can make a distributed mechanism appear spuriously single-gene.

The integrated Phase 5 conclusion remained conservative: the deletion is best treated as a candidate distributed visual-neural reserve lesion, not a proven single-gene photosensitivity mutation. Broader retinal/RPE dysfunction, developmental/network modification, and stress-dependent vulnerability are better supported than photic seizure causation.

4. Final Evidence Adjudication

Claim	Phase 6 status	Evidence boundary
The CNV is structurally real and biologically non-neutral	Supported	Genomic lesion established; multiple constituent loci have dosage/perturbational consequences.
Retinal/RPE mechanisms are	Strongly supported	Mechanism, expression opportunity,

relevant		and constituent-gene perturbation converge; exact-CNV retinal phenotype remains unmeasured.
Membrane/endosomal and stress-response reserve may be reduced	Supported as a mechanistic model	Multiple loci converge, but combined heterozygous function has not been directly measured.
Neurodevelopment/network organization may be altered	Plausible and partially supported	Strong-model evidence exists; heterozygous full-CNV circuit phenotype is not established.
Multigene synergy or epistasis is present	Not demonstrated	Convergence is not equivalent to interaction, synergy, or epistasis.
The retina has increased photon-response gain	Unsupported by current record	Available physiology more often points toward reduced output, slower recovery, or vulnerability.
The CNV causes abnormal V1 flicker entrainment or cortical propagation	Unresolved / not demonstrated	Direct genotype-specific visual-cortical physiology is absent.
The CNV causes a photoparoxysmal response or visually triggered seizures	Insufficient evidence	The causal chain stops before the disease-specific physiological endpoint.

5. Critical Assessment of the Work

5.1 What the project did well

It froze a precise genomic reference and resisted syndrome-level overgeneralization from larger 17p13.3 deletions containing different genes.

It enforced an evidence hierarchy: deletion -> dosage -> molecular perturbation -> spatial opportunity -> cellular/developmental effect -> visual physiology -> phenotype -> causality.

It distinguished genotype states rather than treating knockout, biallelic loss, pathogenic variants, and heterozygous deletion as interchangeable.

It preserved counterevidence and null findings, including compensation, normal heterozygous states, reduced rather than increased response, and failure to identify magnocellular or V1-specific signatures.

It progressively narrowed the working hypothesis instead of escalating claim strength as more literature accumulated.

Phase 5 explicitly addressed reproducibility hazards such as low independent replicate count, pseudoreplication, multiple-frequency testing, and model simplification.

5.2 Where the methodology remains vulnerable

Literature dependence. Much of the mechanistic evidence is assembled across different species, tissues, ages, perturbations, and assay types. Biological coherence can be real while still failing to transfer quantitatively to this exact CNV.

Sparse exact-genotype evidence. Most loci lack direct measurements in cells carrying the complete heterozygous deletion. This is the largest limitation in converting mechanism into CNV-specific biology.

Potential ascertainment and confirmation bias. The project began from a visual-neurological question, which creates a risk that visual or seizure-relevant literature was more deeply sampled than unrelated phenotypic domains. The counterevidence framework reduces but cannot fully remove that risk.

Pathway and network inflation. Gene-dense CNVs can appear convergent because broadly expressed genes participate in common cellular systems. Computational convergence cannot substitute for direct combinatorial perturbation.

Transcript and breakpoint uncertainty. The array-defined boundaries do not determine the exact chromosome junction, transcript fate, nonsense-mediated decay, promoter/enhancer effects, or allele-specific compensation for INPP5K and RTN4RL1.

Expression-resource limitations. Single-cell and single-nucleus atlases have dropout, donor, sampling, batch, regional-dissection, developmental-stage, and annotation constraints; expression cannot establish dysfunction.

Model dependence. Even disciplined ordinal or deterministic evidence models depend on chosen weights, equivalence penalties, and missingness rules. Robustness analyses help identify sensitivity but cannot convert sparse biology into calibrated causal probability.

5.3 Adversarial interpretation

A skeptical reading can explain much of the current record without invoking photosensitivity: the CNV may be biologically consequential because it affects several broadly important cellular systems, while the apparent visual-neural convergence may reflect a mixture of genuine retinal relevance, broad neuronal expression, and literature availability. That null/weak-specificity alternative has not been eliminated. Conversely, the persistence of retinal/RPE, trafficking, stress-response, and network-development signals across independent evidence types argues against treating the interval as biologically irrelevant. The balanced conclusion is therefore neither “causal” nor “nothing”: the deletion remains a credible candidate reserve lesion whose disease-specific physiological consequences are unmeasured.

6. Evidence Boundary and Falsifiability

The highest supported causal level is: structurally defined heterozygous CNV with selected demonstrated dosage consequences, mechanistically relevant retinal/neural pathways, spatiotemporal opportunity, constituent-gene molecular/cellular phenotypes, and distributed functional convergence. The project does not cross the required bridge into exact-CNV retinal physiology, thalamic transfer, V1 flicker physiology, cortical inhibitory containment, pathological frequency entrainment, long-range propagation, PPR, or visually triggered seizure causation.

Evidence that would materially weaken the functional-reserve model includes: normal molecular and physiological behavior of an isogenic full-CNV heterozygous system across the relevant retinal/neural assays under adequate power; complete compensation of predicted dosage-sensitive nodes; absence of stress-dependent differences after calibrated challenge; or demonstration that matched genomic control intervals show equal or stronger domain-specific convergence under the same analytical pipeline. Evidence that would strengthen the model requires exact-genotype perturbation and, ideally, rescue.

7. Path A - Continue In Silico Research with Mentorship and Peer Exposure

This path treats the first-pass record as a reproducible research portfolio to be stress-tested by domain experts rather than as a finished causal claim. The near-term objective is methodological exposure: place the frozen pipeline, evidence tables, assumptions, and negative findings in front of mentors or peers in human genetics, retinal biology, neurophysiology, bioinformatics, or epilepsy research and invite attempts to reproduce, criticize, or break the model.

Convert the Phase 1-5 evidence base into a clean public or mentor-reviewable repository with frozen inputs, source manifest, scripts, environment specification, and explicit expected outputs.

Ask a mentor or peer to independently audit a bounded component first - for example the CNV structural map, RPE convergence, genotype-equivalence coding, dLGN/V1 interpretation, or Phase 5 statistical assumptions.

Reanalyze one or two public datasets from raw or minimally processed data under preregistered questions, rather than adding more narrative literature synthesis.

Seek external review of the null model and the phenotype-ascertainment problem: whether an unrelated gene-dense CNV would appear similarly “convergent” under the same workflow.

Present negative and contradictory findings alongside positive results. The project should gain credibility by surviving criticism, not by maximizing the number of supportive mechanisms.

Use mentor feedback to define a Version 2 research protocol. Do not alter this frozen Version 1 record; any changes should be explicit amendments or a new project pass.

A useful endpoint for Path A would be a mentor-reviewed methods note, reproducible notebook/repository, poster, undergraduate research presentation, or small methods manuscript that focuses on evidence integration and rare-CNV hypothesis discipline rather than claiming a new clinical syndrome.

8. Path B - Transition Toward Wet-Lab Research

8.1 Brief research proposal

Provisional title: Functional Validation of a Heterozygous 17p13.3 Microdeletion as a Distributed Retinal-Neural Reserve Lesion.

Central question: Does the exact heterozygous chr17:1,411,408-1,841,103 deletion produce measurable, rescue-sensitive deficits in retinal/RPE maintenance, membrane/endosomal biology, stress response, neural development, or visually driven network dynamics?

Primary hypothesis: The CNV produces modest, distributed deficits in retinal/RPE and neural functional reserve that may be compensated at baseline but become measurable under controlled physiological or stress challenge. Photosensitivity is a downstream exploratory phenotype, not the primary assumed outcome.

Alternative/null hypotheses: (1) the heterozygous CNV is substantially compensated and produces no reproducible phenotype under the tested conditions; (2) one locus dominates the phenotype; (3) phenotypes are developmental/network-specific without retinal involvement; or (4) retinal abnormalities occur without abnormal cortical flicker propagation.

8.2 Specific aims

Create or obtain isogenic wild-type, heterozygous full-CNV, corrected/rescued, and selected single-gene heterozygous human cellular lines. Confirm CNV structure and transcript/protein dosage before phenotyping.

Test retinal/RPE reserve using retinal organoids and/or RPE models. Priority endpoints: phosphoinositide pools, lipid composition, outer-segment handling, lysosomal/endosomal function, oxidative/DNA-damage response, dark-adaptation or electrophysiologic analogues, and recovery after calibrated challenge.

Test neural/network reserve in relevant human neuronal systems. Priority endpoints: progenitor behavior, neurite morphology, synapse density, E/I-associated measures, calcium dynamics, multielectrode-array firing/bursting, frequency entrainment, phase locking, and long-range propagation.

Use genotype correction or targeted rescue to distinguish CNV-specific effects from background variation and to identify whether observed phenotypes are distributed or locus-dominant.

8.3 Minimal experimental design principles

Use independent biological clones/donors; do not treat many cells from one clone as independent replication.

Predefine primary endpoints and flicker-frequency analyses; control family-wise or false-discovery error across frequency scans and multiple phenotypes.

Blind image/trace quantification where practicable and preserve raw data.

Report null results and directionally contradictory effects.

Power studies for modest heterozygous effects rather than knockout-scale effects.

Treat rescue as a major causal discriminator.

8.4 Funding and fellowship strategy

The most realistic transition is mentor- and institution-based. A trainee should first join or collaborate with a laboratory that already has the required cell culture, retinal organoid/RPE, electrophysiology, imaging, or network-analysis infrastructure. The project can then be adapted to the host laboratory rather than attempting to acquire an entire platform de novo.

NIH/NEI training mechanisms such as F31 or F32 are appropriate once the investigator is in the corresponding predoctoral or postdoctoral training stage and working with an eligible mentor/institution.

NIH R15/AREA mechanisms can support small-scale biomedical projects at eligible undergraduate-focused institutions, but the principal investigator must satisfy institutional and

appointment requirements; this is better viewed as a mentor/institutional route than an immediate independent application.

Vision-focused foundations can be relevant when the proposal is framed around retinal cell biology or inherited retinal mechanisms; epilepsy foundations become more appropriate only if the project directly tests seizure/network physiology rather than relying on indirect mechanistic relevance.

The grant pitch should emphasize the unusual strength of the preliminary in silico work as a hypothesis-narrowing and assay-prioritization package, while explicitly stating that the central physiological claims remain unvalidated. The strongest proposal is one that funds the missing measurements, not one that presents the missing measurements as already solved.

9. Recommended Decision Point

The two paths are complementary rather than mutually exclusive. The immediate low-cost continuation is Path A: external methodological scrutiny, replication, and mentorship. Path B becomes appropriate when a host laboratory, mentor, and assay platform can support exact-genotype experiments. The first wet-lab study should be deliberately narrower than the full project and should target one or two mechanistic bottlenecks with high discriminating value.

A particularly efficient first experiment would compare isogenic wild-type, full-CNV heterozygous, and corrected cells in a retinal/RPE model under baseline and calibrated stress. If no reproducible molecular or physiological deficit is detected with adequate power, the distributed reserve hypothesis should be weakened. If a deficit is detected and rescued, the project gains a direct CNV-specific causal foothold before moving toward more complex thalamic/cortical photic models.

10. Final Project Conclusion

The first-pass in silico program establishes a structurally precise, biologically meaningful 17p13.3 heterozygous deletion containing multiple loci with credible dosage, retinal, membrane, trafficking, stress-response, and neurodevelopmental relevance. Across the project, the evidence repeatedly converged on reduced or altered functional reserve rather than on a single-gene or simple hyperexcitable-retina mechanism. The strongest phenotype-level support concerns retinal/RPE maintenance, photoreceptor integrity, intracellular material handling, stress resilience, and neural-development/network organization. General epileptogenic modification remains plausible but incompletely genotype-matched. Epileptic photosensitivity remains downstream of an unmeasured physiological gap.

Accordingly, the project does not conclude that the CNV causes photosensitivity, a photoparoxysmal response, or visually triggered seizures. It concludes that the CNV is a credible candidate distributed visual-neural reserve lesion whose most important unresolved questions now require either independent methodological scrutiny or direct experimental perturbation. This narrowing is the principal scientific accomplishment of the first pass.

11. Phase 6 Freeze Declaration

PHASE 6 - COMPLETE

The first-pass in silico Genome Research project is hereby frozen as Version 1.0 as of 30 September 2026. The controlling structural reference remains GRCh37/hg19 chr17:1,411,408-1,841,103, nominally 429,696 bp, heterozygous. The frozen interpretive record is that the deletion is biologically consequential and mechanistically consistent with a distributed visual-neural functional-reserve model, while direct causation of epileptic photosensitivity by the exact CNV remains unresolved.

Future research must not silently modify this record. New evidence should be handled by one of three mechanisms: (1) an erratum correcting a demonstrable factual or computational error; (2) a formally reopened analysis that states which frozen conclusion is being reconsidered and why; or (3) a Version 2 research program that preserves Version 1 and documents all changes in data, assumptions, methods, and interpretation.

Null findings, contradictory evidence, failed hypotheses, and unresolved measurements are part of the frozen scientific record and must remain preserved.

12. Current Forward-Looking Funding Notes (30 September 2026)

These items are examples of funding structures relevant to future continuation and are not claims of present applicant eligibility. Eligibility must be checked at the time of application.

National Eye Institute: F31 predoctoral and F32 postdoctoral fellowship mechanisms support mentored vision research training; F31 support can extend up to five years and F32 generally supports two to three years.

NIH R15/AREA: designed for eligible undergraduate-focused institutions; current NIH guidance states that the PD/PI must hold a primary appointment at the eligible organization and that projects may run up to three years. Current NIH pages list total direct-cost limits up to \$375,000 for the applicable AREA mechanism.

Foundation Fighting Blindness: the FY27 schedule lists an Individual Investigator Research Award LOI deadline of 8 October 2026, a Clinical/Research Fellowship deadline of 15 October 2026, a Career Development Award deadline of 28 January 2027, and a Postdoctoral Fellowship LOI deadline of 8 October 2026. These programs are oriented toward retinal disease research and require program-specific eligibility and scope fit.

CURE Epilepsy: Taking Flight supports early-career epilepsy investigators and requires defined postdoctoral-level eligibility and mentor support. Any application from this project should wait until the work contains a direct epilepsy/network physiology aim rather than using retinal or molecular plausibility as a surrogate.

Record of Source Documents

Phase 1: Locus Characterization and Structural Definition of the 17p13.3 Microdeletion - Deletion Map v1.0.

Phase 2: Molecular and Chemical Characterization of an Interstitial 17p13.3 Microdeletion - Comprehensive Synthesis.

Phase 3: Expression, Cellular Localization, Visual-System Distribution, and Developmental Timing; associated Phase 3 research and methods records.

Phase 4: Dosage Sensitivity, Molecular Perturbation, Cellular Phenotype, Developmental Perturbation, Functional Convergence, Visual-System Relevance, and Integrated Adjudication - Frozen Master Report v1.0.

Phase 5: Deterministic Evidence Integration, In-Silico Gap Resolution, Mechanistic Modeling, and Reproducibility Analysis - Frozen scientific record and casual-reader summary v1.0.

Funding Sources Consulted for Forward-Looking Notes

National Eye Institute, Funding for Training and Career Development, accessed 30 September 2026.

NIH Grants & Funding, Research Enhancement Awards (R15), accessed 30 September 2026.

Foundation Fighting Blindness, Scientific Grants and Awards Programs / FY27 schedule, accessed 30 September 2026.

CURE Epilepsy, Grants Program and Opportunities for Epilepsy Researchers, accessed 30 September 2026.